

8 (**short**) An open topped cart is moving at velocity v along a horizontal path, without any friction. It is raining, and the cart is filled by a mass Ω of water each second, as shown in Fig. 6.

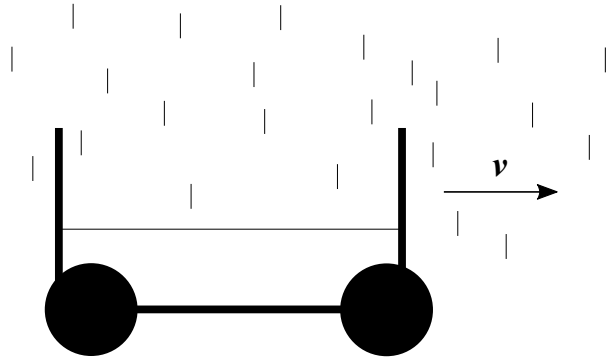


Fig. 6

- (a) Find the horizontal force F that must be applied to the cart to maintain its velocity. [4]
- (b) Find the power required to maintain the cart's velocity. [3]
- (c) Find the efficiency with which this power is converted to kinetic energy. Where does the balance of the energy go? [3]